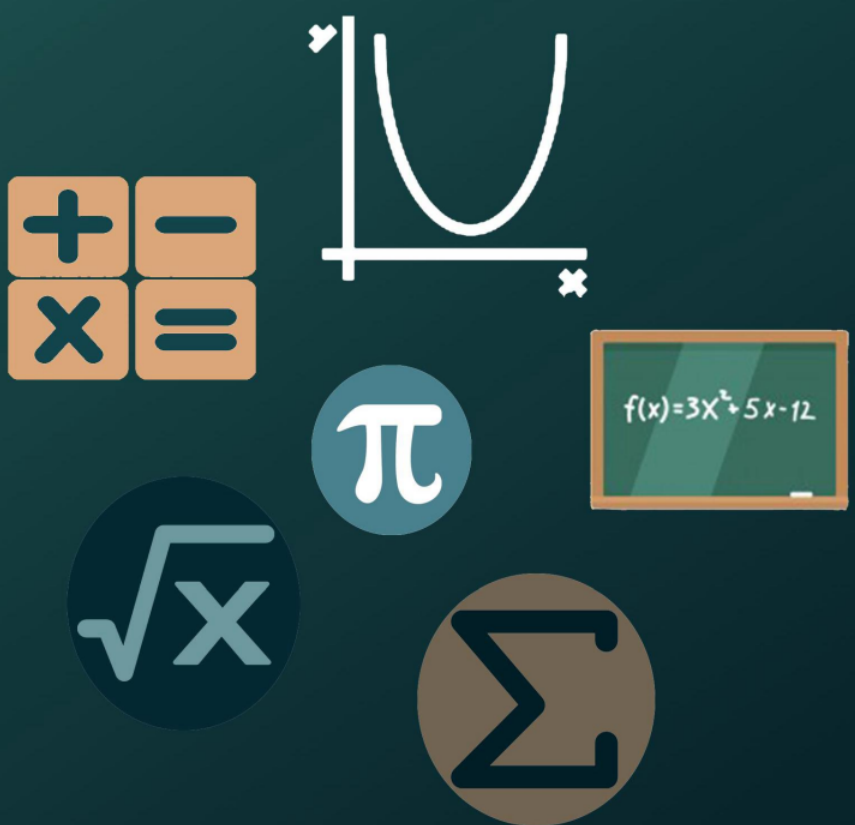
MATHEMATICS

Vector Algebra



SHEET

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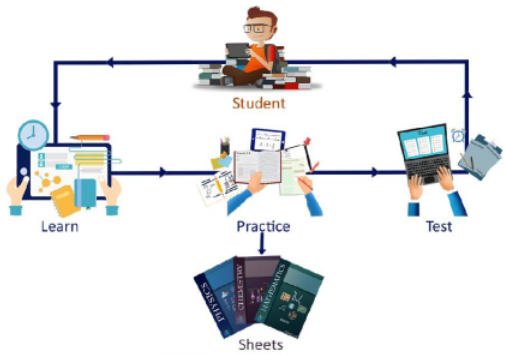
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Note

Detailed solutions are available on the eSaral App.

Module Description



For mastering the concepts only learning is not sufficient. We have to practice and apply those concepts in problem solving. This sheet does just that. It contains a collection of problems segregated in the following exercises to help you master the concepts in a systematic and organized way.

"Practice makes a man perfect"

1. Concept builder – 1 & 1A

As soon as you have finished learning the concept do the problems from these exercises first. These exercises contains easy level questions to help you build your concepts.

1 ⟶ Contains Single Correct Type questions

1A ⟶ Contains pattern based questions incorporating the latest JEE Advanced based patterns like more than one correct, matching list, match the column, etc.

2. Brain Booster – 2 & 2A

Now that you have built your concepts it's time to master them by solving Brain Boosting problems. Don't hurry through these problems. Take time to solve & learn from them. These exercises contains Medium & Tough level problems.

Do questions from 2 & 2A after attempting 1 & 1A

2 ⟶ Single Correct Type questions.

2A ⟶ Pattern Based questions.

3. Simulator – JM & JA

Contains questions from previous year JEE Mains & JEE Advanced questions in exercise JM & exercise JA respectively. Get the real taste & feel of the type of questions being asked in JEE. It's a great tool for simulating your mind with JEE problems.

These exercises are not included in the module but are provided separately.

JM ⟶ JEE Mains previous years topic wise questions.

JA ⟶ JEE Advance previous years topic wise questions.

Home Work Index

Problem solving is an integral part of learning.

Find questions to solve after each video in the homework Index. Make sure that you attempt all the problems (in Ex 1 to 2A) after learning a topic from the videos. For example if you have finished topic 8, first attempt all the problems listed in the index corresponding to topic 8 before proceeding to the video of topic 9.

Sr. No.	Topics	Ex-1	Ex-1A	Ex-2	Ex-2A
1	General Definitions	1-3	23		
2	Vector Addition	4-7		1-2	
3	Geometry of Vectors	8-10	24		1,15
4	Linear Combination of Vectors	11,12		3	2, 18
5	Test of Collinearity	13-15		4	
6	Scalar Product (Dot Product)	16-21	1-2, 20, 25, 26	5-11	3-4, 9-11
7	Vector Product (Cross Product)	22-25	21	12-15	
8	Vector Equation of a Line	26	3-5		
9	Vector Equation of the Angle Bisectors	27	10-13	16	23
10	Fundamental Theorem in Plane		27		5
11	Shortest Distance between Two Skew Lines				19
12	Product of Three or More Vectors	28-37	14-18	17-23	17,20, 21,24
13	Vector Triple Product	38-40	6, 28	24	8, 12-14, 25
14	Scalar Product of Four Vectors				7, 16
15	Coplanarity of Four Points		22, 29		
16	Reciprocal System of Vectors	41-43			6
17	Solving Vector Equations		19		26
18	After Completing Chapter		30, 7-9	25,26	22,27

Exercise - 1 Concept Builder

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

General Definitions

Q1. The position vector of a point C w.r.t. B is $(\hat{i}+\hat{j})$ and that of B w.r.t A is $(\hat{i}-\hat{j})$. The position vector of C w.r.t A is-

(A) $2\hat{i}$ (B) $-2\hat{j}$

(C) $2\hat{j}$ (D) $-2\hat{j}$

Q2. Let D, E, F be the middle points of the sides BC, CA, AB respectively of a triangle ABC.

Then, $\overrightarrow{AD}+\overrightarrow{BE}+\overrightarrow{CF}$ equals-

(A) $\vec{0}$

(B) 0

(C) 2

(D) None of these

Q3. The non-zero vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$ are related $\vec{a}=8\vec{b}$ and $\vec{c}=-7\vec{b}$. Then the angle between $\vec{a}$ and $\vec{c}$ is-

(A) 0 (B) $\pi/4$

(C) $\pi/2$ (D) π

Vector Addition

Q4. If $\vec{a},\vec{b}$ are represented by the sides AB and BC of a regular hexagon ABCDEF, then vector represented by FA will be-

(A) $\vec{a}+\vec{b}$ (B) $\vec{b}-\vec{a}$

(C) $\vec{a}-\vec{b}$ (D) $2\vec{b}-\vec{a}$

Q5. If $r_1=2\hat{i}+4\hat{j}-5\hat{k}$ and $r_2=\hat{i}+2\hat{j}+3\hat{k}$, then the unit vector parallel to the resultant of r_1 and r_2 is-

(A) $3\hat{i}+6\hat{j}-2\hat{k}$

(B) $\hat{i}+\hat{j}-\hat{k}$

(C) $(3\hat{i}+6\hat{j}-2\hat{k})/7$

(D) $(\hat{i}+\hat{j}-\hat{k})/7$

Q6. If C is the mid point of AB and P is any point outside AB, then-

(A) $\overrightarrow{PA}+\overrightarrow{PB}=2\overrightarrow{PC}$

(B) $\overrightarrow{PA}+\overrightarrow{PB}=\overrightarrow{PC}$

(C) $\overrightarrow{PA}+\overrightarrow{PB}+2\overrightarrow{PC}=\vec{0}$

(D) $\overrightarrow{PA}+\overrightarrow{PB}+\overrightarrow{PC}=\vec{0}$

Q7. The co-ordinates of the point in which the line joining the points (3, 5, –7) and (–2, 1, 8) is intersected by the plane *yz* are given by :

(A) $\left(0,-\frac{13}{5},-2\right)$ (B) $\left(0,-\frac{13}{5},\frac{2}{5}\right)$

(C) $\left(0,\frac{13}{5},\frac{2}{5}\right)$ (D) $\left(0,\frac{13}{5},2\right)$

Geometry of Vectors

Q8. The centroid of the triangle whose vertices are $\hat{i}+2\hat{j}$, $2\hat{i}+\hat{j}$, $\hat{i}+\hat{j}+\hat{k}$ is-

(A) $4\hat{i}+4\hat{j}+\hat{k}$ (B) $\frac{4\hat{i}+4\hat{j}+\hat{k}}{3}$

(C) $\frac{4\hat{i}+4\hat{j}+\hat{k}}{2}$ (D) None of these

Q9. If D,E,F are mid points of sides BC, CA, and AB respectively of a triangle ABC, and $\hat{i}+\hat{j}$, $\hat{j}+\hat{k}$, $\hat{k}+\hat{i}$ are p.v. of points A,B and C respectively, then p.v. of centroid of ΔDEF is-

(A) $\frac{\hat{i}+\hat{j}+\hat{k}}{3}$ (B) $\hat{i}+\hat{j}+\hat{k}$

(C) $2(\hat{i}+\hat{j}+\hat{k})$ (D) $\frac{2(\hat{i}+\hat{j}+\hat{k})}{3}$

Q10. Let $\vec{p}$ is the p.v. of the orthocentre & $\vec{g}$ is the p.v. of the centroid of the triangle ABC where circumcentre is the origin. If $\vec{p}=K\vec{g}$, then K =

(A) 3 (B) 2

(C) 1/3 (D) 2/3

Linear Combination of Vectors

Q11. If vectors $2\hat{i}-\hat{j}+k$, $\hat{i}+2\hat{j}-3k$ and $3\hat{i}+a\hat{j}+5\hat{k}$ are coplanar, then the value of a is -

(A) 2 (B) –2

(C) –1 (D) –4

Q12. A vector $\vec{a}$ has components 2p & 1 with respect to a rectangular cartesian system. The system is rotated through a certain angle about the origin in the counterclockwise sense.

If with respect to the new system, $\vec{a}$ has components p+1 & 1 then -

(A) p = 0

(B) p = 1 or p = – 1/3

(C) p = – 1 or p = 1/3

(D) p = 1 or p = – 1

Test of Collinearity

Q13. If 2a – 3b, b and a – b are position vectors of three points A,B and C then they are-

(A) Collinear (B) Non-collinear

(C) Can't say anything (D) None of these

Q14. If $\hat{i}+2\hat{j}+3\hat{k}$ is parallel to sum of the vectors $3\hat{i}+\lambda\hat{j}+2\hat{k}$ and $-2\hat{i}+3\hat{j}+\hat{k}$, then λ equals to-

(A) 1 (B) –1

(C) 2 (D) –2

Q15. If $\vec{A}=(x+1)\,\vec{a}+(2y-3)\,\vec{b}$ and $\vec{B}=5\vec{a}-2\vec{b}$ are two vectors such that $2\vec{A}=3\vec{B}$, & $\vec{a},\vec{b}$ are non collinear vectors then-

(A) x = 13/2, y = 0 (B) x = 0, y = 3

(C) x = –13/2, y = 0 (D) None of these

Scalar Product (Dot Product)

Q16. If vectors $3\hat{i}+2\hat{j}+8\hat{k}$ and $2\hat{i}+x\hat{j}+\hat{k}$ are perpendicular then x is equal to-

(A) 7 (B) –7

(C) 5 (D) –4

Q17. If $\vec{a}=\hat{i}+3\hat{j}-2\hat{k}$ and $\vec{b}=4\hat{i}-2\hat{j}+4\hat{k}$ then $(2\vec{a}+\vec{b}).(\vec{a}-2\vec{b})$ equals to-

(A) 14 (B) –14

(C) 0 (D) –7

Q18. If $\vec{P}=\hat{i}+\hat{j}+\hat{k}$ and $\vec{Q}=\hat{i}+\hat{j}-\hat{k}$ and $\vec{a}$ and $\vec{b}$ are two vectors such that $\vec{P}=2\vec{a}+\vec{b}$ and $\vec{Q}=\vec{a}+2\vec{b}$, then angle between $\vec{a}$ and $\vec{b}$ will be-

(A) $\cos^{-1}\left(\frac{7}{9}\right)$ (B) $\cos^{-1}\left(\frac{7}{11}\right)$

(C) $\cos^{-1}\left(-\frac{7}{11}\right)$ (D) $\cos^{-1}\left(\frac{6\sqrt{2}}{11}\right)$

Q19. If the vectors $\hat{i}-2x\hat{j}-3y\hat{k}$ and $\hat{i}+3x\hat{j}+2y\hat{k}$ are orthogonal to each other, then the locus of the point (x,y) will be-

(A) A circle (B) An ellipse
(C) A parabola (D) A straight line

Q20. The projection of vector $\hat{i}-2\hat{j}+\hat{k}$ on the vector $4\hat{i}-4\hat{j}+7\hat{k}$ is-

(A) 19 (B) 19/4
(C) 9 (D) 19/9

Q21. Given two vectors $\vec{a}=2\hat{i}-3\hat{j}+6\hat{k}$, $\vec{b}=-2\hat{i}+2\hat{j}-\hat{k}$ and $\lambda=\frac{\text{the projection of } \vec{a} \text{ on } \vec{b}}{\text{the projection of } \vec{b} \text{ on } \vec{a}}$, then the value of λ is

(A) $\frac{3}{7}$ (B) 7 (C) 3 (D) $\frac{7}{3}$

Vector Product (Cross Product)

Q22. If $\vec{a}=2\hat{i}+2\hat{j}-\hat{k}$ and $\vec{b}=6\hat{i}-3\hat{j}+2\hat{k}$ then $\vec{a}\times\vec{b}$ equals-

(A) $2\hat{i}-2\hat{j}-\hat{k}$ (B) $\hat{i}-10\hat{j}-18\hat{k}$
(C) $\hat{i}+\hat{j}+\hat{k}$ (D) $6\hat{i}-3\hat{j}+2\hat{k}$

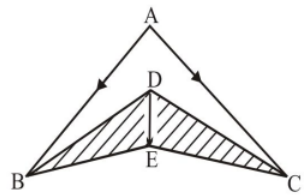
Q23. Let the vectors $\vec{a}$ and $\vec{b}$ such that $|\vec{a}|=3$ and $|\vec{b}|=\frac{\sqrt{2}}{3}$, the $\vec{a}\times\vec{b}$ is a unit vector, Then the angle between $\vec{a}$ and $\vec{b}$ is-

(A) $\frac{\pi}{6}$ (B) $\frac{\pi}{4}$
(C) $\frac{\pi}{3}$ (D) $\frac{\pi}{2}$

Q24. For any vector $\vec{a}$, the value of $(\vec{a}\times\hat{i})^2+(\vec{a}\times\hat{j})^2+(\vec{a}\times\hat{k})^2$ is equal to-

(A) $3\vec{a}^2$ (B) $\vec{a}^2$
(C) $2\vec{a}^2$ (D) $4\vec{a}^2$

Q25. Let $\overrightarrow{AB}=3\hat{i}-\hat{j}$, $\overrightarrow{AC}=2\hat{i}+3\hat{j}$ and $\overrightarrow{DE}=4\hat{i}-2\hat{j}$. The area of the shaded region in the adjacent figure, is -



(A) 5 (B) 6
(C) 7 (D) 8

Vector Equation of a Line

Q26. The projection of the vector $\hat{i}+\hat{j}+\hat{k}$ on the line $\vec{r}=3\hat{i}-\hat{j}+\lambda(\hat{i}+2\hat{j}+3\hat{k})$ is -

(A) $\frac{1}{\sqrt{14}}$ (B) 6
(C) $\frac{6}{\sqrt{14}}$ (D) None of these

Vector Equation of the Angle Bisectors

Q27. Let $\overrightarrow{OA}=\hat{i}+3\hat{j}-2\hat{k}$ and $\overrightarrow{OB}=3\hat{i}+\hat{j}-2\hat{k}$. The vector $\overrightarrow{OC}$ bisecting the angle AOB where C is a point on the line AB is -

(A) $2(\hat{i}+\hat{j}-\hat{k})$ (B) $4(\hat{i}+\hat{j}-\hat{k})$
(C) $\hat{i}+\hat{j}+\hat{k}$ (D) None of these

Product of Three or More Vectors

Q28. If $\vec{a}=4i-3j+k$, $\vec{b}=3i+2j-k$ and $\vec{c}=3i-j+2k$ represent three coterminous edges of a parallelopiped then its volume is-

(A) 60 (B) 15
(C) 30 (D) 40

Q29. The vectors $\hat{i}+3\hat{j}$, $5\hat{k}$ and $p\hat{i}-\hat{j}$, are coplanar, then the value of p is -

(A) 3 (B) –3

(C) 1/3 (D) –1/3

Q30. Given vectors $\vec{x}=3\hat{i}-6\hat{j}-\hat{k}$, $\vec{y}=\hat{i}+4\hat{j}-3\hat{k}$ and $\vec{z}=3\hat{i}-4\hat{j}-12\hat{k}$ then the projection of $\vec{x}\times\vec{y}$ on vector $\vec{z}$ is-

(A) 14 (B) –14

(C) 12 (D) 15

Q31. If the vertices of any tetrahedron is $\vec{a}=\hat{j}+2\hat{k}, \vec{b}=3\hat{i}+\hat{k}, \vec{c}=4\hat{i}+3\hat{j}+6\hat{k}$ and $\vec{d}=2\hat{i}+3\hat{j}+2\hat{k}$ then find its volume.

(A) 36 (B) 6

(C) 12 (D) 24

Q32. Let $\vec{a}=2\hat{i}+3\hat{j}-\hat{k}$ and $\vec{b}=\hat{i}-2\hat{j}+3\hat{k}$. Then the value of λ for which the vector $\vec{c}=\lambda\hat{i}+\hat{j}+(2\lambda-1)\hat{k}$ is parallel to the plane containing $\vec{a}$ and $\vec{b}$, is-

(A) 1 (B) 0

(C) –1 (D) 2

Q33. Let a, b and c be distinct non-negative numbers. If the vectors $a\hat{i}+a\hat{j}+c\hat{k}$, $\hat{i}+\hat{k}$ and $c\hat{i}+c\hat{j}+b\hat{k}$ lie in a plane, then c is-

(A) The Geometric Mean of a and b

(B) The Arithmetic Mean of a and b

(C) Equal to zero

(D) The Harmonic Mean of a and b

Q34. If $\vec{a}$, $\vec{b}$, $\vec{c}$ are non-coplanar vectors and λ is a real number then $[\lambda(\vec{a}+\vec{b}) \quad \lambda^2\vec{b} \quad \lambda\vec{c}]=[\vec{a} \quad \vec{b}+\vec{c} \quad \vec{d}]$ for.

(A) exactly one value of λ

(B) no value of λ

(C) exactly three values of λ

(D) exactly two values of λ

Q35. Let $\vec{a}=\hat{i}-\hat{k}$, $\vec{b}=x\hat{i}+\hat{j}+(1-x)\hat{k}$ and $\vec{c}=y\hat{i}+x\hat{j}+(1+x-y)\hat{k}$. Then $[\vec{a}, \vec{b}\ \vec{c}]$ depends on-

(A) Only y (B) Only x

(C) Both x and y (D) Neither x nor y

Q36. Let $\vec{a}=\hat{i}+\hat{j}+\hat{k}$, $\vec{b}=\hat{i}-\hat{j}+2\hat{k}$ and $\vec{c}=x\hat{i}+(x-2)\hat{j}-\hat{k}$. If the vector $\vec{c}$ lies in the plane of $\vec{a}$ and $\vec{b}$, then x equals-

(A) 0 (B) 1 (C) – 4 (D) –2

Q37. $(\vec{a}+2\vec{b}-\vec{c})\cdot\{(\vec{a}-\vec{b})\times(\vec{a}-\vec{b}-\vec{c})\}$ is equal to-

(A) $[\vec{a}, \vec{b}\ \vec{c}]$ (B) $2[\vec{a}, \vec{b}\ \vec{c}]$

(C) $3[\vec{a}, \vec{b}\ \vec{c}]$ (D) 0

Vector Triple Product

Q38. If $\vec{a}=\hat{i}+2\hat{j}-2\hat{k}$, $\vec{b}=2\hat{i}-\hat{j}+\hat{k}$ and $\vec{c}=\hat{i}+3\hat{j}-\hat{k}$ then $(\vec{a}\times\vec{b}\times\vec{c})$ is equal to-

(A) $20\hat{i}-3\hat{j}+7\hat{k}$ (B) $20\hat{i}-3\hat{j}-7\hat{k}$

(C) $20\hat{i}+3\hat{j}-7\hat{k}$ (D) None of these

Q39. Let $|\vec{a}|=|\vec{b}|=1$ and $|\vec{a}+\vec{b}|=\sqrt{3}$ and also $\vec{c}$ is a vector satisfiying the condition $\vec{c}-\vec{a}-2\vec{b}=3(\vec{a}\times\vec{b})$, then $\vec{c}.\vec{b}$ will be-

(A) –1/2 (B) 1/2

(C) 3/2 (D) 5/2

Q40. If $(\vec{a}\times\vec{b})\times\vec{c}=\vec{a}\times(\vec{b}\times\vec{c})$, where $\vec{a}$, $\vec{b}$ and $\vec{c}$ are any three vectors such that $\vec{a}\cdot\vec{b}\neq 0$, $\vec{b}\cdot\vec{c}\neq 0$, then $\vec{a}$ and $\vec{c}$ are-

(A) inclined at an angle of $\pi/6$ between them

(B) perpendicular

(C) parallel

(D) inclined at an angle of $\pi/3$ between them

Reciprocal System of Vectors 16

Q41. If $\vec{a}, \vec{b}, \vec{c}$ are three non-coplanar vectors, then $\frac{\vec{a}.\vec{b}\times\vec{c}}{\vec{c}\times\vec{a}.\vec{b}}+\frac{\vec{b}.\vec{a}\times\vec{c}}{\vec{c}.\vec{a}\times\vec{b}}$ equals-

(A) 0 (B) 2

(C) $2[\vec{a}, \vec{b}, \vec{c}]$ (D) None of these

Q42. If $\vec{p}=\frac{b\times c}{[abc]}, \vec{q}=\frac{c\times a}{[abc]}, \vec{r}=\frac{a\times b}{[abc]}$, where $\vec{a}, \vec{b}, \vec{c}$ are non-coplanar vectors, then $(\vec{a}+\vec{b}+\vec{c})\cdot(\vec{p}+\vec{q}+\vec{r})$ equals :-

(A) 3 (B) 2

(C) 1 (D) 0

Q43. If $\vec{a}, \vec{b}, \vec{c}$ are three non-coplanar vectors and $\vec{p}, \vec{q}, \vec{r}$ are vectors defined as:

$\vec{p}=\frac{\vec{b}\times\vec{c}}{[\vec{a}\,\vec{b}\,\vec{c}]}, \vec{q}=\frac{\vec{c}\times\vec{a}}{[\vec{a}\,\vec{b}\,\vec{c}]}, \vec{r}=\frac{\vec{a}\times\vec{b}}{[\vec{a}\,\vec{b}\,\vec{c}]}$ then

$(\vec{a}+\vec{b}).\vec{p}+(\vec{b}+\vec{c}).\vec{q}+(\vec{c}+\vec{a}).\vec{r}$ equals-

(A) 0 (B) 1

(C) 2 (D) 3

Exercise - 1A *Concept Builder*

ONE OR MORE THAN ONE

CORRECT TYPE

*Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. Let $\vec{a}=2\hat{i}-\hat{j}+\hat{k}, \vec{b}=\hat{i}+2\hat{j}-\hat{k}$ and $\vec{c}=\hat{i}+\hat{j}-2\hat{k}$ be three vectors. A vector in the plane of $\vec{b}$ and $\vec{c}$ whose projection on $\vec{a}$ is magnitude $\sqrt{2/3}$ is

(A) $2\hat{i}+3\hat{j}-3\hat{k}$

(B) $2\hat{i}+3\hat{j}+3\hat{k}$

(C) $-2\hat{i}-5\hat{j}+\hat{k}$

(D) $2\hat{i}+\hat{j}+5\hat{k}$

Q2. $\vec{a}, \vec{b}, \vec{c}$ are mutually perpendicular vectors of equal magnitude then angle between $\vec{a}+\vec{b}+\vec{c}$ and $\vec{a}$ is -

(A) $\cos^{-1}\left(\frac{1}{3}\right)$ (B) $\cos^{-1}\left(\frac{1}{\sqrt{3}}\right)$

(C) $\pi-\cos^{-1}\left(\frac{1}{\sqrt{3}}\right)$ (D) $\tan^{-1}\sqrt{2}$

Q3. Vector equation of a line passing through points (1, 0, – 1) and (2, 5, 1) is :-

(A) $\hat{i}-\hat{k}+\lambda(\hat{i}+5\hat{j}+2\hat{k})$

(B) $2\hat{i}+5\hat{j}+\hat{k}+\lambda(\hat{i}-\hat{k})$

(C) $2\hat{i}+5\hat{j}+\hat{k}+\lambda(\hat{i}+5\hat{j}+2\hat{k})$

(D) $\hat{i}-\hat{k}+\lambda(2\hat{i}+5\hat{j}+\hat{k})$

Q4. Which of the following statement(s) is/are true in respect of the lines $\vec{r}=\vec{a}+\lambda\vec{b};\ \vec{r}=\vec{c}+\mu d$; where $\vec{b}\times\vec{d}\neq 0$?

(A) Acute angle between the lines is $\cos^{-1}\left(\frac{\vec{b}.\vec{d}}{|\vec{b}||\vec{d}|}\right)$

(B) The lines would intersect if $[\vec{c}\ \vec{b}\ \vec{d}]=[\vec{a}\ \vec{b}\ \vec{d}]$

(C) The lines will be skew if $[\vec{c}-\vec{a}\ \vec{b}\ \vec{d}]\neq 0$

(D) If the lines intersect at $\vec{r}=\vec{r}_0$ then the equation of the plane containing the lines is $[\vec{r}-\vec{r}_0\ \vec{b}\ \vec{d}]=0$.

Q5. Taken on side $\overrightarrow{AC}$ of a triangle ABC, a point M such that $\overrightarrow{AM}=\frac{1}{3}\overrightarrow{AC}$. A point N is taken on the side $\overrightarrow{CB}$ such that $\overrightarrow{BN}=\overrightarrow{CB}$ then, for the point of intersection X of $\overrightarrow{AB}$ & $\overrightarrow{MN}$ which of the following holds good ?

(A) $\overrightarrow{XB}=\frac{1}{3}\overrightarrow{AB}$ (B) $\overrightarrow{AX}=\frac{1}{2}\overrightarrow{AB}$

(C) $\overrightarrow{XN}=\frac{3}{4}\overrightarrow{MN}$ (D) $\overrightarrow{XM}=3\overrightarrow{XN}$

Q6. Let $\vec{a},\vec{b},\vec{c}$ are non-zero vectors such that they are not orthogonal pairwise and such that $\vec{V}_1=\vec{a}\times(\vec{b}\times\vec{c})$ and $\vec{V}_2=(\vec{a}\times\vec{b})\times\vec{c}$. If $\vec{V}_1=\vec{V}_2$ then which of the following hold(s) good?

(A) $\vec{a}$ and $\vec{b}$ are orthogonal

(B) $\vec{a}$ and $\vec{c}$ are collinear

(C) $\vec{b}$ and $\vec{c}$ are orthogonal

(D) $\vec{b}=\lambda(\vec{a}\times\vec{c})$ when λ is a scalar

Q7. If $\vec{a},\vec{b},\vec{c},\vec{d},\vec{e},\vec{f}$ are position vectors of 6 points A, B, C, D, E & F respectively such that $3\vec{a}+4\vec{b}=6\vec{c}+\vec{d}=4\vec{e}+3\vec{f}=\vec{x}$, then -

(A) $\overrightarrow{AB}$ is parallel to $\overrightarrow{CD}$

(B) line AB, CD and EF are concurrent

(C) $\frac{\vec{x}}{7}$ is position vector of the point dividing CD in ratio 1 : 6

(D) A, B, C, D, E & F are coplanar

Q8. $\vec{a}$ and $\vec{b}$ are mutually perpendicular unit vectors. $\vec{r}$ is a vector satisfying $\vec{r}.\vec{a}=0$, $\vec{r}.\vec{b}=1$ and $[\vec{r}\ \vec{a}\ \vec{b}]=1$, then $\vec{r}$ is -

(A) $\vec{a}+(\vec{a}\times\vec{b})$

(B) $\vec{b}+(\vec{a}\times\vec{b})$

(C) $\vec{a}+\vec{b}(\vec{a}\times\vec{b})$

(D) $\vec{a}-\vec{b}+(\vec{a}\times\vec{b})'$

Q9. Read the following statement carefully and identify the true statement -

(i) Two lines parallel to a third line are parallel.

(ii) Two lines perpendicular to a third line are parallel.

(iii) Two lines parallel to a plane are parallel.

(iv) Two lines perpendicular to a plane are parallel.

(v) Two lines either intersect or are parallel.

(A) i & ii (B) i & iv

(C) iv & v (D) i

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph # 1 (Ques. 10 to 13)

Let A(1, 2, 3), B(0, 0, 1), C(– 1, 1, 1) are the vertices of a ΔABC.

Q10. The equation of internal angle bisector through A to side BC is :

(A) $\vec{r} = \hat{i} + 2\hat{j} + \hat{k} + \mu(3\hat{i} + 2\hat{j} + 3\hat{k})$

(B) $\vec{r} = \hat{i} + 2\hat{j} + 3\hat{k} + \mu(3\hat{i} + 4\hat{j} + 3\hat{k})$

(C) $\vec{r} = \hat{i} + 2\hat{j} + 3\hat{k} + \mu(3\hat{i} + 3\hat{j} + 2\hat{k})$

(D) $\vec{r} = \hat{i} + 2\hat{j} + 3\hat{k} + \mu(3\hat{i} + 3\hat{j} + 4\hat{k})$

Q11. The equation of altitude through B to side AC is :

(A) $\vec{r} = \hat{k} + t(7\hat{i} - 10\hat{j} + 2\hat{k})$

(B) $\vec{r} = \hat{k} + t(-7\hat{i} + 10\hat{j} + 11\hat{k})$

(C) $\vec{r} = \hat{k} + t(7\hat{i} - 10\hat{j} - 2\hat{k})$

(D) $\vec{r} = \hat{k} + t(7\hat{i} + 10\hat{j} + 2\hat{k})$

Q12. The equation of median through C to side AB is:

(A) $\vec{r} = -\hat{i} + \hat{j} + \hat{k} + p(3\hat{i} - 2\hat{k})$

(B) $\vec{r} = -\hat{i} + \hat{j} + \hat{k} + p(3\hat{i} + 2\hat{k})$

(C) $\vec{r} = -\hat{i} + \hat{j} + \hat{k} + p(-3\hat{i} + 2\hat{k})$

(D) $\vec{r} = -\hat{i} + \hat{j} + \hat{k} + p(-3\hat{i} - 2\hat{k})$

Q13. The area (ΔABC) is equal to:

(A) $\frac{9}{2}$ (B) $\frac{\sqrt{17}}{2}$

(C) $\frac{17}{2}$ (D) $\frac{7}{2}$

Paragraph # 2 (Ques. 14 to 16)

Three points A(1, 1, 4), B(0, 0, 5) & C(2, –1, 0) forms a plane. P is a point lying on the line $\vec{r} = \hat{i} + 3\hat{j} + \lambda(\hat{i} + \hat{j} + \hat{k})$. The perpendicular distance of point P from plane ABC is $\frac{2\sqrt{6}}{3}$. 'Q' is a point inside the tetrahedron PABC such that resultant of vectors $\overrightarrow{AQ}$, $\overrightarrow{BQ}$, $\overrightarrow{CQ}$ & $\overrightarrow{PQ}$ is a null vector.

Q14. Co-ordinates of point 'P' is -

(A) (2, 4, 1)

(B) (1, 3, 0)

(C) (4, 6, 3)

(D) (7, 9, 6)

Q15. Volume of tetrahedron PABC is -

(A) $\frac{4\sqrt{81}}{9}$ (B) $\frac{2\sqrt{81}}{9}$

(C) $\frac{\sqrt{81}}{9}$ (D) $\frac{6\sqrt{81}}{9}$

Q16. Co-ordinates of point 'Q' is -

(A) $\left(\frac{5}{4}, 1, \frac{5}{2}\right)$ (B) (5, 1, 5)

(C) $\left(\frac{5}{2}, 1, \frac{5}{4}\right)$ (D) $\left(\frac{5}{4}, 5, \frac{5}{2}\right)$

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q17. Match the area/volume of Column - 1 and Column - II

Column - I	**Column - II**
(A) Traingle with vertices $(-\hat{i}+2\hat{j}+3\hat{k})$, $(2\hat{i}-\hat{j}-\hat{k})$, $(\hat{i}+\hat{j}-\hat{k})$	(P) $\frac{\sqrt{3}}{2}$
(B) Tetrahadron with vertices (0, 0, 0), (1,1,– 1), (1, – 1, 1), (– 1, 1, 1)	(Q) $\frac{2}{3}$
(C) Tetrahedron with vertices (–1, 0, 1), (2, – 1, 0), (1, 2, 5), (1, 2, 1)	(R) $\frac{\sqrt{89}}{2}$
(D) Triangle with vertices (1, 0, 0), (0, 1, 0), (1, 1, 1)	(S) $\frac{16}{3}$

Q18. Match the statements / expressions given in Column–I with the values given in Column–II :-

Column–I	**Column–II**
(A) Volume of the parallelepiped with its edges represented by the vectors $\hat{i}+\hat{j}$, $\hat{i}+2\hat{j}$ and $\hat{i}+\hat{j}+\pi\hat{k}$	(P) $\frac{\pi}{6}$
(B) Angle between vectors $\vec{a}$ and $\vec{b}$ where $\vec{a},\vec{b}$ and $\vec{c}$ are unit vectors satisfying $\vec{a}+\vec{b}+\sqrt{3}\vec{c}=\vec{0}$	(Q) $\frac{\pi}{4}$
(C) Angle between the vectors $\vec{a}=\sqrt{3}\hat{i}+\hat{k}$ and $\vec{b}=\hat{i}$	(R) $\frac{\pi}{3}$
	(S) π

Q19. Let $\hat{a},\hat{b}$ and $\hat{c}$ are three unit vectors mutually inclined to each other at an angle θ . $\hat{d}$ is also a unit vector such that $\hat{d}.\hat{a}=\hat{d}.\hat{b}=\hat{d}.\hat{c}=\cos\alpha$.

Match the entries for θ given in column-I with their corresponding values for a given in column-II

Column-I (θ)	**Column-II** (α)
(A) $\frac{\pi}{2}$	(P) $\frac{\pi}{4}$
(B) $\frac{\pi}{3}$	(Q) $\cos^{-1}\left(\frac{1}{\sqrt{3}}\right)$
(C) $\cos^{-1}\left(\frac{1}{4}\right)$	(R) $\cos^{-1}\sqrt{\frac{2}{3}}$
(D) $\cos^{-1}\left(\frac{5}{8}\right)$	(S) $\frac{\pi}{6}$

NUMERICAL TYPE

*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

Q20. Let $\vec{A}=\hat{i}-2\hat{j}+3\hat{k}$, $\vec{B}=2\hat{i}+\hat{j}-\hat{k}$, $\vec{C}=\hat{j}+\hat{k}$. If the vector $\vec{B}\times\vec{C}$ can be expressed as a linear combination $\vec{B}\times\vec{C}=x\vec{A}+y\vec{B}+z\vec{C}$ where x,y,z are scalars, then find the value of $(100x + 10y + 8z)$.

Q21. Let $\vec{a}=-\hat{i}-\hat{k}, \vec{b}=-\hat{i}+\hat{j}$ and $\vec{c}=\hat{i}+2\hat{j}+3\hat{k}$ and three given vectors. If $\vec{r}$ is a vector such that $\vec{r}\times\vec{b}=\vec{c}\times\vec{b}$ and $\vec{r}.\vec{a}=0$, then the value of $\vec{r}.\vec{b}$ is

Q22. If the four points $2\vec{a}+3\vec{b}-\vec{c}$, $\vec{a}-2\vec{b}+3\vec{c}$, $3\vec{a}+4\vec{b}-2\vec{c}$ and $\vec{a}-\lambda\vec{b}+6\vec{c}$ are coplanar then find the value of λ.

SUBJECTIVE TYPE

Q23. If position vectors of A and B are $2\hat{i}+3\hat{j}+4\hat{k}$ and $3\hat{i}-4\hat{j}-5\hat{k}$ then find $\overrightarrow{AB}$.

Q24. Find the position vector of the point which divide the join of the points $2\vec{a}-3\vec{b}$ and $3\vec{a}-2\vec{b}$ internally and externally in the ratio 2 : 3.

Q25. $\vec{a},\vec{b},\vec{c}$ and $\vec{d}$ are the position vectors of the points A ≡ (x, y, z); B ≡ (y, –2z, 3x) ; C ≡ (2z, 3x, –y) and D ≡ (1, –1, 2) respectively. If $|\vec{a}|=2\sqrt{3};(\vec{a}\wedge\vec{b})=(\vec{a}\wedge\vec{c});(\vec{a}\wedge\vec{d})=\frac{\pi}{2}$ and $(\vec{a}\wedge\hat{j})$ is obtuse, then find x, y, z.

Q26. In a unit cube. Find

(a) The angle between the diagonal of the cube and a diagonal of a face skew to it.

(b) The angle between the diagonals of two faces of the cube through the same vertex.

(c) The angle between a diagonal of a cube and a diagonal of a face intersecting it.

Q27. The base vectors $\vec{a}_1,\vec{a}_2,\vec{a}_3$ are given in terms of base vectors $\vec{b}_1,\vec{b}_2,\vec{b}_3$ as $\vec{a}_1=2\vec{b}_1+3\vec{b}_2-\vec{b}_3$; $\vec{a}_2=\vec{b}_1-2\vec{b}_2+2\vec{b}_3$ & $\vec{a}_3=-2\vec{b}_1+\vec{b}_2-2\vec{b}_3$. If $\vec{F}=3\vec{b}_1-\vec{b}_2+2\vec{b}_3$, then express $\vec{F}$ in terms of $\vec{a}_1,\vec{a}_2$ & $\vec{a}_3$.

Q28. If $\vec{a}=\hat{i}+\hat{j}-\hat{k}$, $\vec{b}=-\hat{i}+2\hat{j}+2\hat{k}$ & $\vec{c}=-\hat{i}+2\hat{j}-\hat{k}$, find a unit vectors normal to the vectors $\vec{a}+\vec{b}$ and $\vec{b}-\vec{c}$.

Q29. The vector $\overrightarrow{OP}=\hat{i}+2\hat{j}+2\hat{k}$ turns through a right angle, passing through the positive x-axis on the way. Find the vector in its new position.

Q30. The pv's of the four angular points of a tetrahedron are $A(\hat{j}+2\hat{k})$; $B(3\hat{i}+\hat{k})$; $C(4\hat{i}+3\hat{j}+6\hat{k})$ & $D(2\hat{i}+3\hat{j}+2\hat{k})$.

Find :

(i) The perpendicular distance from A to the line BC.

(ii) The volume of the tetrahedron ABCD.

(iii) The perpendicular distance from D to the plane ABC.

(iv) The shortest distance between the lines AB & CD.

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Q1. Let two non-collinear unit vectors $\hat{a}$ and $\hat{b}$ form an acute angle. A point P moves so that at any time t the position vector $\overrightarrow{OP}$ (where O is the origin) is given by $\hat{a}\cos t + \hat{b}\sin t$. When P is farthest from origin O, let M be the length of $\overrightarrow{OP}$ and $\hat{u}$ be the unit vector along $\overrightarrow{OP}$.

Then -

(A) $\hat{u} = \dfrac{\hat{a}+\hat{b}}{|\hat{a}+\hat{b}|}$ and $M = (1+\hat{a}.\hat{b})^{1/2}$

(B) $\hat{u} = \dfrac{\hat{a}-\hat{b}}{|\hat{a}-\hat{b}|}$ and $M = (1+\hat{a}.\hat{b})^{1/2}$

(C) $\hat{u} = \dfrac{\hat{a}+\hat{b}}{|\hat{a}+\hat{b}|}$ and $M = (1+2\hat{a}.\hat{b})^{1/2}$

(D) $\hat{u} = \dfrac{\hat{a}-\hat{b}}{|\hat{a}-\hat{b}|}$ and $M = (1+2\hat{a}.\hat{b})^{1/2}$

Q2. If ABCDEF is a regular hexagon and if $\overrightarrow{AB}+\overrightarrow{AC}+\overrightarrow{AD}+\overrightarrow{AE}+\overrightarrow{AF} = \lambda\overrightarrow{AD}$, then λ is -

(A) 0 (B) 1 (C) 2 (D) 3

Q3. If $\vec{a}+5\vec{b}=\vec{c}$ and $\vec{a}-7\vec{b}=2\vec{c}$, then-

(A) $\vec{a}$ and $\vec{c}$ are like but $\vec{b}$ and $\vec{c}$ are unlike vectors

(B) $\vec{a}$ and $\vec{b}$ are unlike vectors and so also $\vec{a}$ and $\vec{c}$

(C) $\vec{b}$ and $\vec{c}$ are like but $\vec{a}$ and $\vec{b}$ are unlike vectors

(D) $\vec{a}$ and $\vec{c}$ are unlike vectors and so also $\vec{b}$ and $\vec{c}$

Q4. Let $\vec{a}$, $\vec{b}$ and $\vec{c}$ be three non-zero vectors such that no two of these are collinear. If the vector $\vec{a}+2\vec{b}$ is collinear with $\vec{c}$ and $\vec{b}+3\vec{c}$ is collinear with $\vec{a}$ (λ being some non-zero scalar) then $\vec{a}+2\vec{b}+6\vec{c}$ equals -

(A) $\lambda\vec{a}$ (B) $\lambda\vec{b}$

(C) $\lambda\vec{c}$ (D) $\vec{0}$

Q5. $\vec{a}$, $\vec{b}$, $\vec{c}$ are three vectors, such that $\vec{a}+\vec{b}+\vec{c}=\vec{0}$, $|\vec{a}|=1$, $|\vec{b}|=2$, $|\vec{c}|=3$, then $\vec{a}.\vec{b}+\vec{b}.\vec{c}+\vec{c}.\vec{a}$ is equal to-

(A) 1 (B) 0

(C) –7 (D) 7

Q6. If $\vec{a}$, $\vec{b}$, $\vec{c}$ are three non-zero, non-coplanar vectors and

$$\vec{b}_1 = \vec{b} - \frac{\vec{b}.\vec{a}}{|\vec{a}|^2}\vec{a},\ \vec{b}_2 = \vec{b} + \frac{\vec{b}.\vec{a}}{|\vec{a}|^2}\vec{a},$$

$$\vec{c}_1 = \vec{c} - \frac{\vec{c}.\vec{a}}{|\vec{a}|^2}\vec{a} + \frac{\vec{b}.\vec{c}}{|\vec{c}|^2}\vec{b}_1,$$

$$\vec{c}_2 = \vec{c} - \frac{\vec{c}.\vec{a}}{|\vec{a}|^2}\vec{a} - \frac{\vec{b}_1.\vec{c}}{|\vec{b}_1|^2}\vec{b}_1,$$

$$\vec{c}_3 = \vec{c} - \frac{\vec{c}.\vec{a}}{|\vec{a}|^2}\vec{a} + \frac{\vec{b}_1.\vec{c}}{|\vec{c}|^2}\vec{b}_1,$$

$$\vec{c}_4 = \vec{c} - \frac{\vec{c}.\vec{a}}{|\vec{a}|^2}\vec{a} - \frac{\vec{b}.\vec{c}}{|\vec{b}|^2}\vec{b}_1$$

then the set of orthogonal vectors is -

(A) $(\vec{a}, \vec{b}_1, \vec{c}_3)$ (B) $(\vec{a}, \vec{b}_1, \vec{c}_2)$

(C) $(\vec{a}, \vec{b}_1, \vec{c}_1)$ (D) $(\vec{a}, \vec{b}_2, \vec{c}_2)$

Q7. Let $\vec{a}=\hat{i}+2\hat{j}+\hat{k}$, $\vec{b}=\hat{i}-\hat{j}+\hat{k}$ and $\vec{c}=\hat{i}+\hat{j}-\hat{k}$.

A vector in the plane of $\vec{a}$ and $\vec{b}$ whose projection on $\vec{c}$ has the magnitude equal to $\dfrac{1}{\sqrt{3}}$ is -

(A) $4\hat{i}-\hat{j}+4\hat{k}$ (B) $3\hat{i}+\hat{j}-3\hat{k}$

(C) $2\hat{i}+\hat{j}-2\hat{k}$ (D) $4\hat{i}+\hat{j}-4\hat{k}$

Q8. Given the points $A(-2, 3, -4)$, $B(3, 2, 5)$, $C(1, -1, 2)$ & $D(3, 2, -4)$. The projection of the vector $\vec{AB}$ on the vector $\vec{CD}$ is -

(A) $\frac{22}{3}$ (B) $-\frac{21}{4}$

(C) $-\frac{47}{7}$ (D) -47

Q9. Two vectors $\vec{p}$, $\vec{q}$ on a plane satisfy $|\vec{p}+\vec{q}| = \sqrt{13}$, $|\vec{p}-\vec{q}| = 1$ and $|\vec{p}| = \sqrt{3}$.

The angle between $\vec{p}$ and $\vec{q}$, is equal to

(A) $\frac{\pi}{6}$ (B) $\frac{\pi}{4}$

(C) $\frac{\pi}{3}$ (D) $\frac{\pi}{2}$

Q10. Let $\vec{u}, \vec{v}, \vec{w}$ be such that $|\vec{u}|=1$, $|\vec{v}|=2$, $|\vec{w}|=3$. If the projection of $\vec{v}$ along $\vec{u}$ is equal to that of $\vec{w}$ along $\vec{u}$ and vectors $\vec{v}$, $\vec{w}$ are perpendicular to each other then $|\vec{u}-\vec{v}+\vec{w}|$ equals:

(A) 2 (B) $\sqrt{7}$

(C) $\sqrt{14}$ (D) 14

Q11. If $\vec{a}, \vec{b}, \vec{c}$ be any three unit vectors such that $3\vec{a}+4\vec{b}+5\vec{c}=0$, then-

(A) $\vec{a} \parallel \vec{b}$ (B) $\vec{b} \parallel \vec{c}$

(C) $\vec{a} \perp \vec{b}$ (D) None of these

Q12. Let $\vec{u}=\hat{i}+\hat{j}$, $\vec{v}=\hat{i}-\hat{j}$ and $\vec{w}=\hat{i}+2\hat{j}+3\hat{k}$. If $\hat{n}$ is a unit vector such that $\vec{u}.\hat{n}=0$ and $\vec{v}.\hat{n}=0$, then $|\vec{w}.\hat{n}|$ is equal to -

(A) 3 (B) 0

(C) 1 (D) 2

Q13. The sine of angle formed by the lateral face ADC and plane of the base ABC of the tetrahedron ABCD where $A \equiv (3, -2, 1)$; $B \equiv (3, 1, 5)$; $C \equiv (4, 0, 3)$ and $D \equiv (1, 0, 0)$ is -

(A) $\frac{2}{\sqrt{29}}$ (B) $\frac{5}{\sqrt{29}}$

(C) $\frac{3\sqrt{3}}{\sqrt{29}}$ (D) $\frac{-2}{\sqrt{29}}$

Q14. Let $\vec{A}$ be vector parallel to line of intersection of planes P_1 and P_2 through origin. P_1 is parallel to the vectors $2\hat{j}+3\hat{k}$ and $4\hat{j}-3\hat{k}$ and P_2 is parallel to $\hat{j}-\hat{k}$ and $3\hat{i}+3\hat{j}$, then the angle between vector $\vec{A}$ and $2\hat{i}+\hat{j}-2\hat{k}$ is -

(A) $\frac{\pi}{6}, \frac{\pi}{2}$ (B) $\frac{\pi}{4}, \frac{\pi}{2}$

(C) $\frac{\pi}{6}, \frac{\pi}{4}$ (D) $\frac{\pi}{4}, \frac{3\pi}{4}$

Q15. Let $\vec{a}, \vec{b}, \vec{c}$ be unit vectors such that $\vec{a}+\vec{b}+\vec{c}=\vec{0}$.

Which one of the following is correct?

(A) $\vec{a}\times\vec{b}=\vec{b}\times\vec{c}=\vec{c}\times\vec{a}=\vec{0}$

(B) $\vec{a}\times\vec{b}=\vec{b}\times\vec{c}=\vec{c}\times\vec{a}\neq\vec{0}$

(C) $\vec{a}\times\vec{b}=\vec{b}\times\vec{c}=\vec{a}\times\vec{c}\neq\vec{0}$

(D) $\vec{a}\times\vec{b}, \vec{b}\times\vec{c}, \vec{c}\times\vec{a}$ are mutually perpendicular

Q16. The vector $\vec{c}$, directed along the internal bisector of angle between the vectors $\vec{a}=7\hat{i}-4\hat{j}-4\hat{k}$ and $\vec{b}=-2\hat{i}-\hat{j}+2\hat{k}$ with $|\vec{c}|=5\sqrt{6}$, is -

(A) $\frac{5}{3}(\hat{i}-7\hat{j}+2\hat{k})$ (B) $\frac{5}{3}(5\hat{i}-5\hat{j}+2\hat{k})$

(C) $\frac{5}{3}(\hat{i}+7\hat{j}+2\hat{k})$ (D) $\frac{5}{3}(-5\hat{i}+5\hat{j}+2\hat{k})$

Q17. A, B, C, D be four points in a space and if, $|\vec{AB} \times \vec{CD} + \vec{BC} \times \vec{AD} + \vec{CA} \times \vec{BD}| = \lambda$ (area of triangle ABC), then the value of λ is -

(A) 4 (B) 2

(C) 1 (D) None of these

Q18. The vector $\vec{a} = \alpha\hat{i} + 2\hat{j} + \beta\hat{k}$, lies in the plane the vectors $\vec{b} = \hat{i} + \hat{j}$ and $\vec{c} = \hat{j} + \hat{k}$, and bisect the angle between $\vec{c}$ and $\vec{b}$. Then which one of the following gives possible values of α and β ?

(A) $\alpha = 2, \beta = 2$ (B) $\alpha = 1, \beta = 2$

(C) $\alpha = 2, \beta = -1$ (D) $\alpha = 1, \beta = 1$

Q19. The number of distinct real values of λ, for which the vectors $-\lambda^2\hat{i} + \hat{j} + \hat{k}$, $\hat{i} - \lambda^2\hat{j} + \hat{k}$ and $\hat{i} + \hat{j} - \lambda^2\hat{k}$ are coplanar, is :-

(A) zero (B) one

(C) two (D) three

Q20. If $\vec{a} = \hat{i} + \hat{j} + \hat{k}$, $\vec{b} = \hat{i} - \hat{j} + \hat{k}$, $\vec{c} = \hat{i} + 2\hat{j} - \hat{k}$, then the value of $\begin{vmatrix} \vec{a}.\vec{a} & \vec{a}.\vec{b} & \vec{a}.\vec{c} \\ \vec{b}.\vec{a} & \vec{b}.\vec{b} & \vec{b}.\vec{c} \\ \vec{c}.\vec{a} & \vec{c}.\vec{b} & \vec{c}.\vec{c} \end{vmatrix} =$

(A) 2 (B) 4

(C) 16 (D) 64

Q21. The edges of a parallelopiped are of unit length and are parallel to non–coplanar unit vectors $\hat{a}, \hat{b}, \hat{c}$ such that $\hat{a}.\hat{b} = \hat{b}.\hat{c} = \hat{c}.\hat{a} = \frac{1}{2}$. Then, the volume of the parallelopiped is :

(A) $\frac{1}{\sqrt{2}}$ (B) $\frac{1}{2\sqrt{2}}$

(C) $\frac{\sqrt{3}}{2}$ (D) $\frac{1}{\sqrt{3}}$

Q22. The altitude of a parallelopiped whose three coterminous edges are the vectors,

$\vec{A} = \hat{i} + \hat{j} + \hat{k}$; $\vec{B} = 2\hat{i} + 4\hat{j} - \hat{k}$

& $\vec{C} = \hat{i} + \hat{j} + 3\hat{k}$ with $\vec{A}$ and $\vec{B}$ as the sides of the base of the parallelopiped, is -

(A) $2/\sqrt{19}$ (B) $4/\sqrt{19}$

(C) $2\sqrt{38}/19$ (D) None

Q23. A variable plane forms a tetrahedron of constant volume 64 K^3 with the coordinate planes and the origin, then locus of the centroid of the tetrahedron is -

(A) $x^3 + y^3 + z^3 = 6K^3$

(B) $xyz = 6k^3$

(C) $x^2 + y^2 + z^2 = 4K^2$

(D) $x^{-2} + y^{-2} + z^{-2} = 4K^{-2}$

Q24. The value of $\hat{i} \times (\vec{r} \times \hat{i}) + \hat{j} \times (\vec{r} \times \hat{j}) + \hat{k} \times (\vec{r} \times \hat{k})$ is -

(A) $\vec{r}$ (B) $2\vec{r}$

(C) $3\vec{r}$ (D) $4\vec{r}$

Q25. Given planes :

$P_1 : cy + bz = x$

$P_2 : az + cx = y$

$P_3 : bx + ay = z$

P_1, P_2 and P_3 pass through one line, if

(A) $a^2 + b^2 + c^2 = ab + bc + ca$

(B) $a^2 + b^2 + c^2 + 2abc = 1$

(C) $a^2 + b^2 + c^2 = 1$

(D) $a^2 + b^2 + c^2 + 2ab + 2bc + 2ca + 2abc = 1$

Q26. If $\vec{A}$, $\vec{B}$ and $\vec{C}$ are three non-coplanar vectors then $(\vec{A} + \vec{B} + \vec{C}).[(\vec{A} + \vec{B}) \times (\vec{A} + \vec{C})]$ equals -

(A) 0 (B) $[\vec{A}\ \vec{B}\ \vec{C}]$

(C) $2[\vec{A}\ \vec{B}\ \vec{C}]$ (D) $-[\vec{A}\ \vec{B}\ \vec{C}]$

ONE OR MORE THAN ONE CORRECT TYPE

*Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. ABCD is a parallelogram. E and F be the middle points of the sides AB and BC, then -

(A) DE divide AC in ratio 2 : 3

(B) DF divide AC in ratio 3 : 2

(C) DE divide AC in ratio 1 : 2

(D) DF divide AC in ratio 2 : 1

Q2. Let $\vec{a}$, $\vec{b}$, $\vec{c}$ are three non-coplanar vectors such that $\vec{r}_1 = \vec{a} - \vec{b} + \vec{c}$, $\vec{r}_2 = \vec{b} + \vec{c} - \vec{a}$, $\vec{r}_3 = \vec{c} + \vec{a} + \vec{b}$, $\vec{r} = 2\vec{a} - 3\vec{b} + 4\vec{c}$.

If $\vec{r} = \lambda_1\vec{r}_1 + \lambda_2\vec{r}_2 + \lambda_3\vec{r}_3$, then -

(A) $\lambda_1 = 7$ (B) $\lambda_1 + \lambda_3 = 3$

(C) $\lambda_1 + \lambda_2 + \lambda_3 = 4$ (D) $\lambda_3 + \lambda_2 = 2$

Q3. A vector which makes equal angles with the vectors $\frac{1}{3}(\hat{i} - 2\hat{j} + 2\hat{k})$, $\frac{1}{5}(-4\hat{i} - 3\hat{k})$, $\hat{j}$ is -

(A) $5\hat{i} + \hat{j} + 5\hat{k}$ (B) $-5\hat{i} + \hat{j} + 5\hat{k}$

(C) $5\hat{i} - \hat{j} - 5\hat{k}$ (D) $5\hat{i} + \hat{j} - 5\hat{k}$

Q4. The vector $\vec{a} = x\hat{i} - 3\hat{j} - \hat{k}$ and $\vec{b} = 2x\hat{i} + x\hat{j} - \hat{k}$ include an acute angle and $\vec{b}$ include an obtuse angle with positive y-axis. Then values of x may be -

(A) –2 (B) – 3

(C) all $x < 0$ (D) all $x > 0$

Q5. If a, b, c are different real numbers and $a\hat{i} + b\hat{j} + c\hat{k}$; $b\hat{i} + c\hat{j} + a\hat{k}$ & $c\hat{i} + a\hat{j} + b\hat{k}$ are position vectors of three non-collinear points A, B & C, then :

(A) centroid of triangle ABC is $\frac{a+b+c}{3}\left(\hat{i} + \hat{j} + \hat{k}\right)$

(B) $\hat{i} + \hat{j} + \hat{k}$ is equally inclined to the three vectors

(C) perpendicular from the origin to the plane of triangle ABC meet at centroid

(D) triangle ABC is an equilateral triangle

Q6. Identify the statement (s) which is/are incorrect?

(A) $\vec{a} \times \left[\vec{a} \times \left(\vec{a} \times \vec{b}\right)\right] = \left(\vec{a} \times \vec{b}\right)\left(\vec{a}^2\right)$

(B) If $\vec{a}$, $\vec{b}$, $\vec{c}$ are non coplanar vectors and $\vec{v}.\vec{a} = \vec{v}.\vec{b} = \vec{v}.\vec{c} = 0$ then $\vec{v}$ must be a null vector

(C) If $\vec{a}$ and $\vec{b}$ lie in a plane normal to the plane containing the vectors $\vec{c}$ and $\vec{d}$ then $\left(\vec{a} \times \vec{b}\right) \times \left(\vec{c} \times \vec{d}\right) = 0$.

(D) If $\vec{a}, \vec{b}, \vec{c}$ and $\vec{a}', \vec{b}', \vec{c}'$ are reciprocal system of vectors then $\vec{a}.\vec{b}' + \vec{b}.\vec{c}' + \vec{c}.\vec{a}' = 3$.

Q7. If $A(\vec{a}); B(\vec{b}); C(\vec{c})$ and $D(\vec{d})$ are four points such that $\vec{a} = -2\hat{i} + 4\hat{j} + 3\hat{k}$; $\vec{b} = 2\hat{i} - 8\hat{j}$; $\vec{c} = \hat{i} - 3\hat{j} + 5\hat{k}$; $\vec{d} = 4\hat{i} + \hat{j} - 7\hat{k}$, d is the shortest distance between the lines AB and CD, then which of the following is true ?

(A) $d = 0$, hence AB and CD intersect

(B) $d = \dfrac{[\overrightarrow{AB}\ \overrightarrow{CD}\ \overrightarrow{BD}]}{|\overrightarrow{AB}\times\overrightarrow{CD}|}$

(C) AB and CD are skew lines and $d = \dfrac{23}{13}$

(D) $d = \dfrac{[\overrightarrow{AB}\ \overrightarrow{CD}\ \overrightarrow{AC}]}{|\overrightarrow{AB}\times\overrightarrow{CD}|}$

Q8. If $\vec{a}$, $\vec{b}$ & $\vec{c}$ are non coplanar unit vectors such that $\vec{a}\times(\vec{b}\times\vec{c}) = \dfrac{\vec{b}+\vec{c}}{\sqrt{2}}$, then the angle between -

(A) $\vec{a}$ & $\vec{b}$ is $\dfrac{3\pi}{4}$ (B) $\vec{a}$ & $\vec{b}$ is $\dfrac{\pi}{4}$

(C) $\vec{a}$ & $\vec{c}$ is $\dfrac{3\pi}{4}$ (D) $\vec{a}$ & $\vec{c}$ is $\dfrac{\pi}{4}$

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph # 1 (Ques. 9 to 11)

If $\vec{b}\neq 0$, then every vector $\vec{a}$ can be written in a unique manner as the sum of a vector $\vec{a}_{11}$ parallel to $\vec{b}$ and a vector $\vec{a}_{\perp}$ perpendicular to $\vec{b}$. If $\vec{a}$ is parallel to $\vec{b}$, then $\vec{a}_{11} = \vec{a}$ and $\vec{a}_{\perp} = 0$. If $\vec{a}$ is perpendicular to $\vec{b}$, then $\vec{a}_{11} = 0$ and $\vec{a}_{\perp} = \vec{a}$. The vector $\vec{a}_{11}$ is called the projection of $\vec{a}$ on $\vec{b}$ and is denoted by $\text{proj}_b\,\vec{a}$. Since $\text{proj}_b\,\vec{a}$ is parallel to $\vec{b}$, it is a scalar multiple of q the unit vector in the direction of $\vec{b}$ i.e. $\text{Proj}_b\,\vec{a} = \lambda\,\vec{\mu}_b$ The scalar λ is called the component of $\vec{a}$ in the direction of $\vec{b}$ and is denoted by $\text{comp}_b\,\vec{a}$. In fact, $\text{Proj}_b\,\vec{a} = (\vec{a}\cdot\vec{u}_b)\vec{u}_b$ and $\text{comp}_b\,\vec{a} = \vec{a}\cdot\vec{u}_b$.

Q9. If $\vec{a} = -2\hat{i}+\hat{j}+\hat{k}$ and $\vec{b} = 4\hat{i}-3\hat{j}+\hat{k}$ then $\text{proj}_b\,\vec{a}$.

(A) $4\hat{i}-3\hat{j}+2\hat{k}$

(B) $-\dfrac{5}{13}(4\hat{i}-3\hat{j}+\hat{k})$

(C) $\dfrac{5}{13}(4\hat{i}-3\hat{j}+\hat{k})$

(D) $-\dfrac{4}{11}(2\hat{i}-\hat{j}+2\hat{k})$

Q10. If $\vec{a}$ and $\vec{b}$ are as in above question and $\vec{a} = \vec{a}_{11}+\vec{a}_{\perp}$ then $\vec{a}_{\perp}$ is equal to -

(A) $\dfrac{1}{13}(-3\hat{i}+\hat{j}+9\hat{k})$

(B) $\dfrac{1}{13}(-3\hat{i}-\hat{j}+4\hat{k})$

(C) $\dfrac{2}{13}(-3\hat{i}-\hat{j}+9\hat{k})$

(D) $-\dfrac{2}{13}(3\hat{i}+\hat{j}-9\hat{k})$

Q11. If $\vec{a}$ and $\vec{b}$ are as in above question then $\text{comp}_b\,\vec{a}$ is -

(A) $(-5/13)\sqrt{26}$ (B) $(5/13)\sqrt{26}$

(C) $5\sqrt{26}$ (D) $-\dfrac{5}{11}\sqrt{17}$

Paragraph # 2 (Ques. 12 to 14)

Consider the three vectors $\vec{p}$, $\vec{q}$ and $\vec{r}$ such that $\vec{p} = \hat{i}+\hat{j}+\hat{k}$, $\vec{q} = \hat{i}-\hat{j}+\hat{k}$, $\vec{p}\times\vec{r} = \vec{q}+c\vec{p}$ and $\vec{p}.\vec{r} = 2$.

Q12. The value of $[\vec{p}\ \vec{q}\ \vec{r}]$ is :

(A) $-\dfrac{5\sqrt{2}c}{|\vec{r}|}$ (B) $-\dfrac{8}{3}$

(C) 0 (D) greater than zero

Q13. If $\vec{x}$ is a vector such that $[\vec{p}\ \vec{q}\ \vec{r}]\vec{x} = (\vec{p}\times\vec{q})\times\vec{r}$, then $\vec{x}$ is :

(A) $c(\hat{i}-2\hat{j}+\hat{k})$

(B) a unit vector

(C) indeterminate

(D) $-\frac{1}{2}(\hat{i}-2\hat{j}+\hat{k})$

Q14. If is a vector satisfying $(1+c)\vec{y} = \vec{p}\times(\vec{q}\times\vec{r})$ then the vectors $\vec{x}, \vec{y}, \vec{r}$.

(A) are collinear

(B) are coplanar

(C) Represent the coterminus edges of a tetrahedron whose volume is 'c' cubic units

(D) Represent the coterminus edges of a parallelepiped whose volume is 'c' cubic units

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q15.

Column I	**Column II**
(A) P is point in the plane of the triangle ABC. pv's of A, B and C are $\vec{a}$, $\vec{b}$ and $\vec{c}$ respectively with respect to P as the origin. If $(\vec{b}+\vec{c})\cdot(\vec{b}-\vec{c})=0$ and $(\vec{c}+\vec{a})\cdot(\vec{c}-\vec{a})=0$, then w.r.t. the triangle ABC, P is its	(P) centroid
(B) If $\vec{a}, \vec{b}, \vec{c}$ are the position vectors of the three non collinear points A, B and C respectively such that the vector $\vec{V} = \overrightarrow{PA}+\overrightarrow{PB}+\overrightarrow{PC}$ is a null vector then w.r.t. the ΔABC, P is its	(Q) orthocentre
(C) If P is a point inside the Δ ABC such that the vector $\vec{R} = (BC)(\overrightarrow{PA})+(CA)(\overrightarrow{PB})+(AB)(\overrightarrow{PC})$ is a null vector then w.r.t. the ΔABC, P is its	(R) Incentre
(D) If P is a point in the plane of the triangle ABC such that the scalar product $\overrightarrow{PA}\cdot\overrightarrow{CB}$ and $\overrightarrow{PB}\cdot\overrightarrow{AC}$ vanishes, then w.r.t. the ΔABC, P is its	(S) circumcentre

Q16. Let $\vec{a}, \vec{b}, \vec{c}$ be any three vectors then match the following :

Column - I	**Column - II**
(A) $\vec{a}\times(\vec{b}+\vec{c})+\vec{b}\times(\vec{c}+\vec{a})$	(P) $[\vec{c}\ \vec{b}\ \vec{a}]\vec{a}$
(B) $\vec{a}\times(\vec{b}\times\vec{c})+\vec{b}\times(\vec{c}\times\vec{a})$	(Q) $[\vec{c}\ \vec{b}\ \vec{a}]\vec{b}$
(C) $(\vec{a}\times\vec{b})\times(\vec{c}\times\vec{b})$	(R) $(\vec{a}\times\vec{b})\times\vec{c}$
(D) $(\vec{a}\times\vec{b})\times(\vec{c}\times\vec{a})$	(S) $(\vec{a}+\vec{b})\times\vec{c}$

TRIPLE MATCH COLUMN TYPE

*This section contains ONE table having 3 columns and 4 rows. Based on the table each question has FOUR options [A], [B], [C] AND [D]. **ONLY ONE** of these four is correct.*

Q17.

Column - I	**Column - II** Volume of parallelopiped whose adjacent edges are are represented by $\vec{a}, \vec{b}$ and $\vec{c}$	**Column - III** Volume of Tetrahedron whose adjacent edges represented by $\vec{a}, \vec{b}$ and $\vec{c}$
(I) $\vec{a} = 2\hat{i} - 3\hat{j} + 4\hat{k}$, $\vec{b} = \hat{i} + 2\hat{j} - \hat{k}$, $\vec{c} = 3\hat{i} - \hat{j} + 2\hat{k}$	(i) $32\sqrt{2}$	(A) $\frac{1}{3}$
(II) $\vec{a} = \hat{i} + \hat{j}$, $\vec{b} = \hat{i} + 2\hat{j}$, $\vec{c} = \hat{i} + \hat{j} + 2\hat{k}$	(ii) 7	(B) $\frac{16}{3}\sqrt{2}$
(III) $\lvert\vec{a}\rvert = \lvert\vec{b}\rvert = \lvert\vec{c}\rvert = 4$, and angle between $\vec{a}$ and $\vec{b}$ is $\pi/3$, angle between $\vec{b}$ and $\vec{c}$ is $\pi/3$ and angle between $\vec{c}$ and $\vec{a}$ is $\pi/3$.	(iii) $8\sqrt{2}$	(C) $\frac{7}{6}$
(IV) $\lvert\vec{a}\rvert = \lvert\vec{b}\rvert = 2, \lvert\vec{c}\rvert = 4$ and angle between $\vec{a}$ and $\vec{b}$ is $\frac{\pi}{2}$, angle between $\vec{b}$ and $\vec{c}$ is $\frac{\pi}{3}$, angle between $\vec{a}$ and $\vec{b}$ and $\vec{c}$ is $\frac{\pi}{3}$, angle between $\vec{a}$ and $\vec{c}$ is $\frac{\pi}{3}$.	(iv) 2	(D) $\frac{4}{3}\sqrt{2}$

NUMERICAL TYPE

*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

Q18. Let $\vec{a},\vec{b},\vec{c}$ are three vectors of which every pair is non collinear, and the vectors $\vec{a}+3\vec{b}$ and $2\vec{b}+3\vec{c}$ are collinear with $\vec{c}$ and $\vec{a}$ respectively. If $\vec{b}.\vec{b}=1$, then find $|2\vec{a}+9\vec{c}|$.

Q19. Let $\vec{a}=\hat{i}-\hat{j}$, $\vec{b}=\hat{i}+2\hat{j}+2\hat{k}$, $\vec{c}=\hat{i}-\hat{j}+\hat{k}$ and $\vec{d}=2\hat{i}-\hat{j}+\hat{k}$. If p is the shortest distance between the lines $\vec{r}=\vec{a}+t\vec{b}$ and $\vec{r}=\vec{c}+p\vec{d}$ find $2p^2$.

Q20. If the vectors $a\hat{i}+\hat{j}+\hat{k}$, $\hat{i}+b\hat{j}+\hat{k}$ and $\hat{i}+\hat{j}+c\hat{k}$, $(a\neq b\neq c\neq 1)$ are coplanar, find the value of $\frac{1}{1-a}+\frac{1}{1-b}+\frac{1}{a-c}$.

Q21. ABCD is a tetrahedron with pv's of its angular point as A(–5, 22, 5); B(1, 2, 3); C(4, 3, 2) and D(–1, 2, –3). If the area of the triangle AEF where the quadrilaterals ABDE and ABCF are parallelograms is $\sqrt{S}$ then find the values of S.

Q22. Given $|\vec{p}|=2$; $|\vec{q}|=3$ and $\vec{p}.\vec{q}=0$.

If $\vec{V}=(\vec{p}\times(\vec{p}\times(\vec{p}\times(\vec{p}\times\vec{q}))))$ if the ratio of $\frac{\vec{V}}{\vec{q}}$ is λ, find $\lambda=$

SUBJECTIVE TYPE

Q23. Incident ray is along the unit vector $\hat{v}$ and the reflected ray is along the unit vector $\hat{w}$. The normal is along unit vector $\hat{a}$ outwards. Express $\hat{w}$ in terms of $\hat{a}$ and $\hat{v}$.

Q24. Let $\vec{a}=\alpha\hat{i}+2\hat{j}-3\hat{k},\vec{b}=\hat{i}+2\alpha\hat{j}-2\hat{k}$ and $\vec{c}=2\hat{i}-\alpha\hat{j}+\hat{k}$. Find the value(s) of α, if any, such that $\{(\vec{a}\times\vec{b})\times(\vec{b}\times\vec{c})\}\times(\vec{c}\times\vec{a})=0$. Find the vector product when $\alpha=0$.

Q25. Prove that $|\vec{a}\times\vec{b}|=\sqrt{-\vec{b}.[\vec{a}\times(\vec{a}\times\vec{b})]}$

Q26. Solve the following equation for the vector $\vec{p}$; $\vec{p}\times\vec{a}+(\vec{p}.\vec{b})\vec{c}=\vec{b}\times\vec{c}$ where $\vec{a},\vec{b},\vec{c}$ are non zero non coplanar vectors and $\vec{a}$ is neither perpendicular to $\vec{b}$ nor to $\vec{c}$, hence show that $\left[\vec{p}\times\vec{a}+\frac{[\vec{a}\,\vec{b}\,\vec{c}]}{\vec{a}.\vec{c}}\vec{c}\right]$ is perpendicular to $\vec{b}-\vec{c}$.

Q27. Points X & Y are taken on the sides QR & RS, respectively of a parallelogram PQRS, so that $\overrightarrow{QX}=4\overrightarrow{XR}$ & $\overrightarrow{RY}=4\overrightarrow{YS}$. The line XY cuts the line PR at Z. Prove that $\overrightarrow{PZ}=\left(\frac{21}{25}\right)\overrightarrow{PR}$.7

Answer Key

Ex-1									
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
A	A	D	C	C	A	D	B	D	A
11	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**
D	B	A	B	A	B	B	C	A	D
21	**22**	**23**	**24**	**25**	**26**	**27**	**28**	**29**	**30**
D	B	B	C	C	C	A	C	D	B
31	**32**	**33**	**34**	**35**	**36**	**37**	**38**	**39**	**40**
B	B	A	B	D	D	C	A	D	C
41	**42**	**43**							
A	A	D							

Ex - 1A

1. (A,C) **2.** (B,D) **3.** (A,C) **4.** (A, B, C, D) **5.** (B,C)

6. (B, D) **7.** (B,C,D) **8.** (B) **9.** (B,D) **10.** (D)

11. (C) **12.** (B) **13.** (B) **14.** (A) **15.** (B)

16. (A)

17. (A)→(R), (B)→(Q), (C)→(S), (D)→(P)

18. (A)→(S), (B)→(R), C→(P)

19. (A) →(Q); (B) → (R); (C) → (P); (D) →(S)

20. 101 **21.** 9 **22.** 6 **23.** $\overrightarrow{AB} = \hat{i} - 7\hat{j} - 9\hat{k}$

24. $-5\vec{b}$

25. x = 2, y = – 2 , z = –2 **26. (a)** $\cot^{-1}(0)$; **(b)** $\cot^{-1}\frac{1}{\sqrt{3}}$; **(c)** $\cot^{-1}\sqrt{2}$

27. $F = 2\vec{a}_1 + 5\vec{a}_2 + 3\vec{a}_3$ **28.** $\pm\hat{i}$

29. $\frac{4}{\sqrt{2}}\hat{i} - \frac{1}{\sqrt{2}}\hat{j} - \frac{1}{\sqrt{2}}\hat{k}$ **30. (i)** $\frac{6}{7}\sqrt{14}$ **(ii)** 6 **(iii)** $\frac{3}{5}\sqrt{10}$ **(iv)** $\sqrt{6}$

Ex-2									
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
A	D	A	D	C	B	A	C	A	C
11	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**
C	A	B	D	B	A	A	D	C	C
21	**22**	**23**	**24**	**25**	**26**				
A	C	B	B	B	D				

Ex - 2A

1. (C,D) **2.** (B,C) **3.** (B,C) **4.** (A,B,C) **5.** (A,B,C,D)

6. (A,C,D) **7.** (B, C, D) **8.** (A,D) **9.** (B) **10.** (C)

11. (A) **12.** (B) **13.** (D) **14.** (C)

15. (A) $\rightarrow$ S; (B) $\rightarrow$ P; (C) $\rightarrow$ R; (D) $\rightarrow$ Q

16. A $\rightarrow$ (S), B $\rightarrow$ (R), C $\rightarrow$ (Q), D $\rightarrow$ (P)

17. (I) - (ii) - (C); (II) - (iv) - (A), (III) - (i) - (B), (IV) - (iii) - (D)

18. 6 **19.** 1 **20.** 1 **21.** 110 **22.** 16

23. $\hat{w} = \hat{v} - 2(\hat{a}.\hat{v})\hat{a}$

24. $\alpha = 2/3$; if $\alpha = 0$ then vector product is $-60(2\hat{i} + \hat{k})$

26. $\left\{\vec{p} = \frac{\left[\vec{a}\,\vec{b}\,\vec{c}\right]}{(\vec{a}.\vec{c})(\vec{a}.\vec{b})}\left(\vec{a} + \vec{c} \times \vec{b}\right) + \frac{(\vec{b}.\vec{c})\,\vec{b}}{(\vec{a}.\vec{b})} - \frac{(\vec{b}.\vec{b})\,\vec{c}}{(\vec{a}.\vec{b})}\right\}$

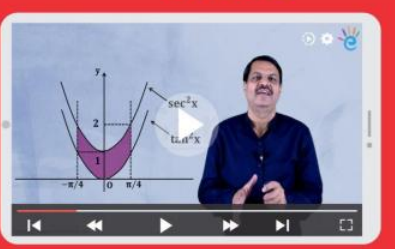
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